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GALACTIC RADIATION AT RADIO FREQUENCIES

II. THE DISCRETE SOURCES

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Summary

An account is given of observations on a number of discrete sources of radio-frequency radiation, together with a brief description of the observational techniques used in their detection, and of the methods employed to determine their positions and angular extent.

Noise spectra of four of the more intense sources have been measured over a frequency range of 40–160 Mc/s.: in three cases the change of intensity with wavelength is found to be greater than that of the background continuum, and in one case less. Two of the sources have been provisionally identified with astronomically rare objects, the Crab Nebula and N.G.C. 5128.

Finally, the short period fluctuations in the intensity of some of the sources, notably that in Cygnus, are discussed. Evidence is presented which, contrary to previous views, suggests that these fluctuations are of terrestrial rather than of extra-terrestrial origin.

I. INTRODUCTION

In 1946, whilst making a survey of galactic noise at 64 Mc/s., Hey, Parsons, and Phillips(1) noticed that the noise intensity from the constellation of Cygnus was by no means constant. They observed fluctuations of about 15 per cent. of the total power received by their aerial system, the average period of the fluctuations being about one minute. By direction finding they localized the source of these fluctuations to an area of 2° centred at Right Ascension 20 hours, Declination +43°, and suspected the existence of other active areas within 8° of this position. Bolton and Stanley(2), using an interference technique with the sea as a reflector, showed that the fluctuations originate in a discrete source less than 6 minutes in angular width, and later reported the existence of further discrete sources(3), apparently constant in intensity. Both teams suggested a possible similarity between the Cygnus fluctuations and enhanced solar radiation associated with active sunspots. Ryle and Smith(4), however, using a vertical interferometer, showed that the radiation from the Cygnus source was unpolarized, in contrast with that from active sunspots. They, too, found a number of other discrete sources.

The first part of this paper discusses the detection of extremely small signals, methods of observing discrete galactic sources, and the determination of their positions and angular widths. The second part gives details of observations on discrete sources made at this Laboratory, including the variation of

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intensity with frequency (over the range 40–160 Mc/s.) for several of the larger sources. Finally, short-period fluctuations in the intensity of the sources are discussed, with particular reference to two years' observations on the Cygnus source, including spaced aerial measurements between Australia and New Zealand.

II. Observational Techniques

(*a*) *Receiver Sensitivity*

Over the range of frequencies considered, galactic noise received on a moderately directive aerial system is of the same order as the noise generated in the receiver. The radiation from individual discrete sources is only a small fraction of the incident noise. The limit of sensitivity of a receiver capable of detecting signals from these small sources is set by the fluctuation level of the total noise observed at the detector. Contributions to the fluctuation level arise from :

(i) Statistical fluctuation of the noise generated in the first stages of the receiver.

(ii) Fluctuations of receiver gain due to variation in filament and high-tension supplies.

(iii) Terrestrial noise.

(i) *Statistical Fluctuations of Noise in the Receiver.*—The theoretical limit of receiver stability is determined by the statistical fluctuations of noise generated in the first stages of the receiver. Dicke(5) has shown that, in terms of the aerial temperature,* this limit is given by the formula

$$\frac{\triangle T}{T}=\frac{\pi^{3/2}N}{8}\left(\frac{\alpha}{\triangle\omega}\right)^{\frac{1}{2}},$$

where N is the noise factor of the receiver,

$\triangle\omega$ is the bandwidth of the receiver (radians per second),

α is the reciprocal of the time constant used in the recording system,

T is the aerial temperature,

$\triangle T$ is the minimum observable temperature change.

For a typical 100 Mc/s. system, where $N=3$, $\alpha=0\cdot2$, and $\triangle\omega=2\pi\times10^6$,

$$\triangle T=0\cdot1\ ^\circ\mathrm{K}.$$

for an aerial at ambient temperature.

(ii) *Supply Voltage Variations.*—It has been found that, with careful attention to the mechanical and electrical layout of the receiver, the only significant changes in receiver output from internal causes are due to variations in gain resulting from fluctuations in the high-tension and filament voltages. For an average 100 Mc/s. receiver a change of 1 part in 3000 in high-tension voltage or 1 part in 1500 in filament voltage produces a change of 1 part in 1000 in second detector voltage. The optimum stability attained in the present

* The concepts *aerial temperature* and *equivalent black-body temperature* have been described in detail in Part I of this series.

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observations was 1 part in 10,000 in high-tension and 1 part in 6000 in filament supplies. For an aerial at ambient temperature this represented a minimum observable temperature change of about 0·3 °K. In the receivers used, the second detector was followed by a direct-current amplifier in which most of the detector output, representing galactic noise plus receiver noise, was balanced out and the remainder presented on a recording milliammeter.

In an alternative system used by Dicke for observation of microwave radiation, the receiver input is switched alternately between the aerial and a matched load at ambient temperature. As the signal is only received for half the time and then integrated, the theoretical limit of sensitivity, as given by the formula quoted above, will be $\sqrt{2}\triangle T$. The practical limits of such a system are unknown. To improve appreciably on the sensitivity of the Dicke system, supply voltages for the previously described system would have to be regulated to the order of 1 part in 10^5.

(iii) *Terrestrial Noise Sources.*—Radio-frequency noise, due to distant atmospherics, car ignition, transmission line corona, etc., sets a practical limit to the sensitivity of any system. In the present investigation the fluctuation level due to terrestrial noise, even during the optimum observing period between 2 a.m. and dawn, was generally greater than that arising within the equipment. Sensitivity close to the practical limit discussed earlier has been achieved in field observations from selected sites.

(*b*) *Interference Technique*

Owing to the difficulty of constructing highly directional aerial systems, discrete sources of radio-frequency energy have only been recognized against the general noise background by using some form of interference technique. Two

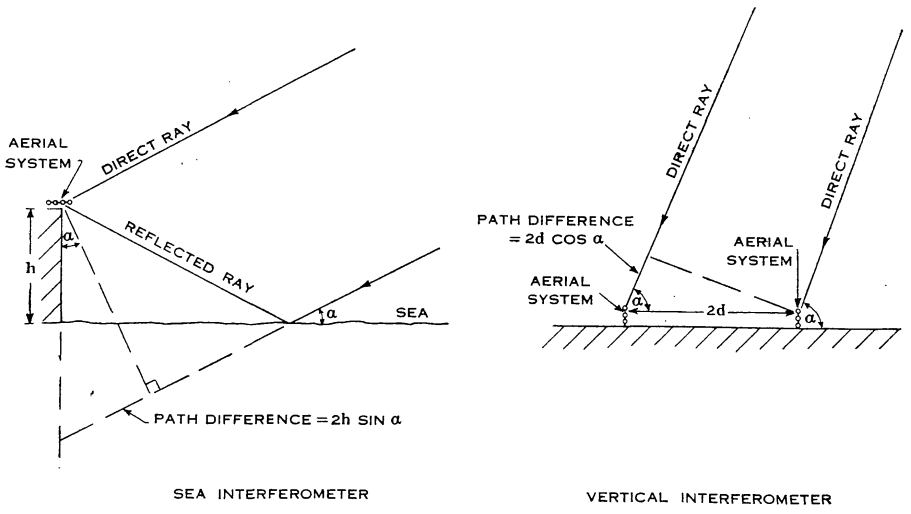


Fig. 1.—Schematic diagram of sea and vertical interferometer systems.

forms of interferometer are in use at present. One, an adaptation of " Lloyd's mirror " using the sea as a reflector, was first used by McCready, Pawsey, and Payne-Scott(6) for studying solar radiation at 200 Mc/s., and later by Bolton and

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Stanley(2) for the detection of discrete sources. The second, similar to the Michelson stellar interferometer and consisting of two spaced aerials, was first used by Ryle and Vonberg(7) for observation of solar radiation at 80 and 175 Mc/s. The principles of the two interferometers are illustrated in Figure 1.

The advantages of the sea interferometer used in the present investigations are :

(i) The same sensitivity is achieved with a single aerial unit.

(ii) The source first appears sharply in contrast to the general background at rising, whereas with the vertical system a slowly increasing amplitude pattern is obtained.

(iii) The sharp appearance at rising is useful in discriminating between a number of sources close together.

(*c*) *Measurement of Angular Width*

It has been shown elsewhere(6, 8) that an estimate of the angular width of the source in a direction at right angles to the horizon can be made from an examination of sea interferometer patterns. If R is the ratio of the received power (above the extrapolated background level) at the maxima and minima of the interference pattern, h the height of the aerial above sea-level, and λ the wavelength used, W the effective angular width is given by

$$W=\frac{\lambda}{\pi h}\sqrt{3R}.$$

In practice, an upper limit is generally set by the noise fluctuation level of the receiver, which affects the measurement of the heights of the minima. It may appear that unlimited resolution can be obtained by merely increasing the height of the observing site, but this is limited by two factors. One is the " effective sea reflection coefficient ", which is less than unity for altitudes below about 1°, due to the earth's curvature. The second is the bandwidth of the receiver, which increases the height of the minima and decreases the height of the maxima with increasing fringe order. Although both these effects can be computed (see Appendices I and II), some uncertainty must remain.

It is obvious that the resolving power of the equipment depends principally on the intensity of the source, and thus accurate measurements have only been made for the more intense sources.

(*d*) *Determination of Position*

For the determination of position in the case of the sea interferometer, each minimum of the patterns obtained at rising or setting represents the time at which the source attains a certain altitude (determined by instrumental and site constants and a knowledge of atmospheric refraction).

The intersection on the celestial sphere of two lines, one of which is determined by the sidereal time of rising and the other by the declination, gives the position of the source. The line of declination is estimated from the periodicity of the interference pattern or by direction finding when an insufficient number of minima is available.

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TABLE 1

LIST OF CONFIRMED SOURCES

Number and Year of Discovery	Temporary Designation	Intensity at 100 Mc/s. (watts m^{-2} (c/s).$^{-1}$) 10^{-24}	Upper Limit of Angular Width (minutes)	Method of Positon Finding	Celestial Coordinates (epoch 1948)	Galactic Coordinates	Number of Observations	Remarks
1—1946	Cygnus-A	125	1·5	R and S	R.A. 19 hr. 58 min. 14 sec. Dec. +40° 36′	l 44° b +4½°	320	Amplitude fluctuation first observed by Hey, Parsons, and Phillips(1)
2—1947	Taurus-A	18·5	6	R and S	R.A. 05 hr. 31 min. 30 sec. Dec. +22° 01′	l 150° b −4°	80	Possibly M.1 (N.G.C. 1952), the Crab Nebula
4—1947	Virgo-A	12·5	5	R and S	R.A. 12 hr. 28 min. 06 sec. Dec. +12° 41′	l 260° b +74°	100	Possibly M.87 (N.G.C. 4486)
5—1947	(Centaurus)	(8)	30				10	See Subsection III (*f*)
6—1947	Centaurus-A	18·5	7	R and S	R.A. 13 hr. 22 min. 20 sec. Dec. −42° 37′	l 274° b +20°	50	Possibly N.G.C. 5128
7—1948	Hercules-A	2	30	R, L of R, D	R.A. 16 hr. 50 min. Dec. +5°	l 351° b +26°	30	See position map I, Fig. 2
8—1948	Taurus-C	3	15	L of R, D	R.A. 04 hr. 38 min. Dec. +28°	l 140° b −10°	15	See position map II, Fig. 2, also Reference (8) for detailed information
9—1948	Taurus-B	6	30	L of R	R.A. 05 hr. 32 min. Dec. +24°	l 151° b −3°	15	See position map II, Fig. 2, also Reference (8) for detailed information
10—1948	Fornax-A	2	15	R, L of R, D	R.A. 03 hr. 11 min. Dec. −36°	l 204° b −57°	15	See position map III, Fig. 2
11—1948	Serpens-Cauda-A	3	15	R, L of R, D	R.A. 18 hr. 43 min. Dec. +5°	l 5° b +11½°	6	See position map IV, Fig. 2
12—1948	(Centaurus)	2	30			l 265° b +17°	4	Rises before Centaurus-A, almost circumpolar at the latitude of the observing site
15—1948	(Leo)	1	30	L of R	R.A. 11 hr. 52 min. Dec. +17°	l 223° b +74°	5	Rises just before Virgo-A. See map V, Fig. 2
17—1949	Scorpius-A	2	30			l 320° b +0°	5	
18—1949	Serpens-Cauda-B	2	30	L of R, D	R.A. 18 hr. 11 min. Dec. −15°	l 343° b −1°	5	See position map VI, Fig. 2
19—1949	Sextans-A	2	30	L of R, D	R.A. 09 hr. 55 min. Dec. −5°	l 214° b +38°	4	See position map VII, Fig. 2
20—1949		2	30	L of R, D	R.A. 05 hr. 01 min. Dec. −36°	l 205° b −35°	10	See position map VIII, Fig. 2. On the border of Colomba and Caelum
21—1949	Puppis-A	3	30	R, L of R, D	R.A. 08 hr. 18 min. Dec. −42°	l 227° b −2°	8	See position map IX, Fig. 2
22—1949	Pictor-A	3	30	R, L of R, D	R.A. 05 hr. 18 min. Dec. −44°	l 217° b −33°	6	See position map X, Fig. 2

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In practice, however, this method is somewhat inaccurate owing to the low reading-accuracy of the times of occurrence of minima and the considerable scatter between results on different occasions (presumably due to refraction). A higher order of accuracy can be obtained if the source is observed at both rising and setting from the same or different sites. This method was used by Bolton and Stanley in determining the position of the source Taurus–A, and is described in detail elsewhere(8).

III. The Discrete Sources

(*a*) *General*

Details of the known discrete sources are set out in Table 1. The larger sources, discovered late in 1947 and early in 1948 with the use of only small aerial systems, have been studied extensively. The smaller sources have been found by use of the large array described in Part I of this series. Owing to the relatively small beam width of this system, systematic surveying is necessarily a lengthy process. As the full sensitivity of the system could only be realized for a few hours before dawn, small sources detected during the optimum observing period could not always be confirmed by later observations. Sources of which less than three records are available have therefore not been included in the table.

Accurate position finding observations have only been made for several of the larger sources. Approximate positions (shown in brackets) have been found by a number of methods designated by the following letters in Table 1 :

R, from observations of the track of the source above the horizon after rising.

R and *S*, from observations of the source at rising and setting.

L of *R*, from the time of rising which determines a line on the celestial sphere.

D, by direction finding at rising.

With the exception of the major sources in Taurus, Virgo, and Centaurus, no great accuracy is claimed for these measurements ; probable limits in the positions of the sources are shown on the series of position maps (Fig. 2).

The angular widths given refer to measurements at 100 Mc/s., as the sensitivity of the equipment was greater at this frequency. Upper limits of 30 minutes of arc are given where no accurate measurements have been made ; it is considered unlikely that such sources would have been seen with the interferometer if their angular widths had exceeded this figure.

Estimates of intensity similarly refer to 100 Mc/s. The accuracy of these measurements varies from ± 10 per cent. for the larger sources to ± 50 per cent. for the very weak sources.

(*b*) *Cygnus–A*

The position of this source, given in Table 1, is the result of a long series of measurements made from the north-east coast of New Zealand in June and July 1948. Almost the complete track of the source above the northern horizon was observed—the reflected ray was intercepted only for a short period between

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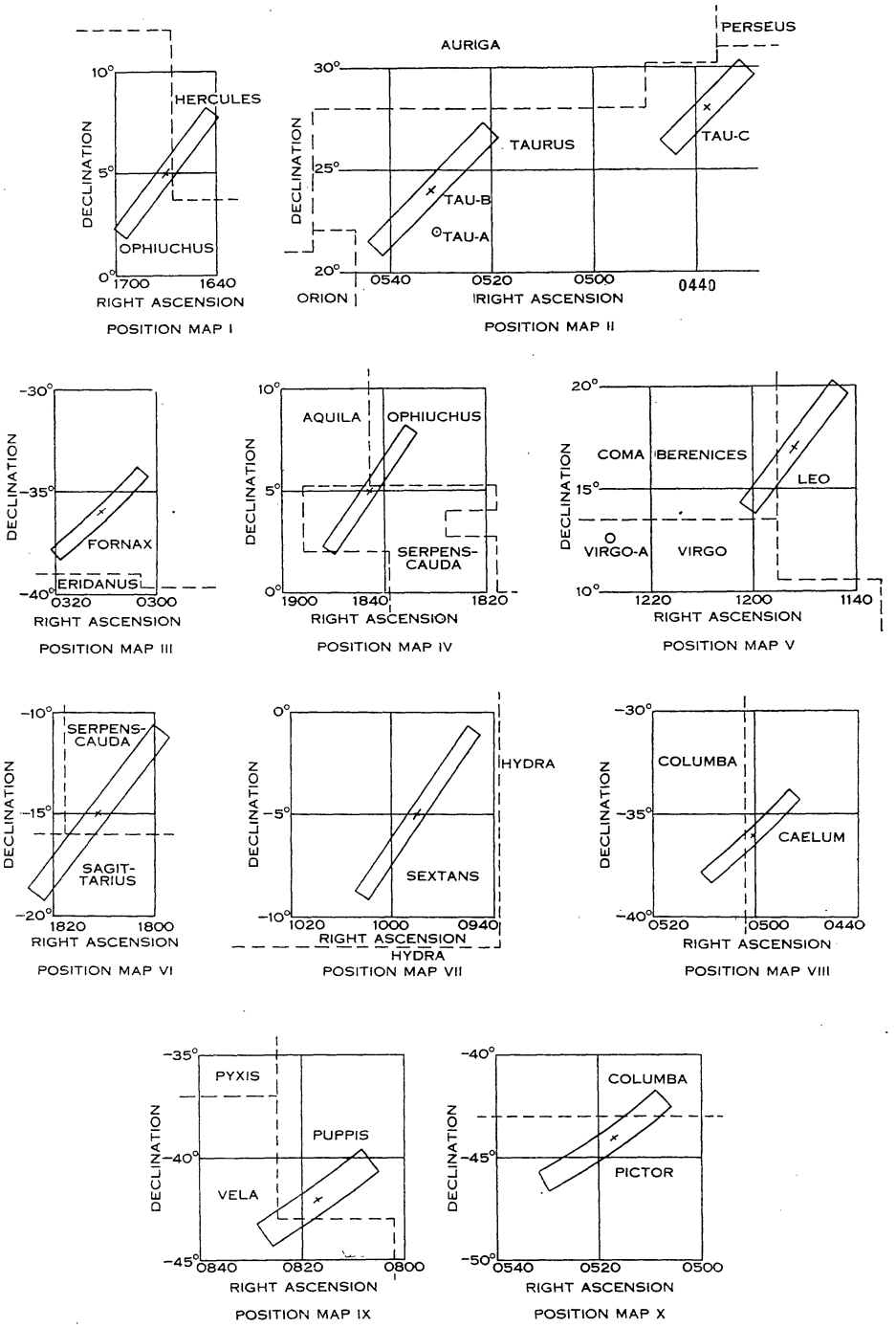


Fig. 2.—Position maps for a number of discrete sources. Full lines mark limits of position and dotted lines boundaries of constellation.

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culmination and setting of the source by a group of islands. The Right Ascension was found from comparison of the time of corresponding minima on each side of culmination and the declination from the semi-diurnal arc and the altitude of the source at culmination. The main feature of these observations was the great inconsistency between the deduced position on different nights and between observations at different periods of the same night. The authors have been unable to reconcile their results with those of Ryle (personal communication) and also of Mills of this Laboratory using vertical interferometers. It is unlikely, however, that the source lies outside the area bounded by Right Ascension 19 hr. 58 min.–20 hr. 00 min., and Declination +40° 20′–40° 40′.

(*c*) *Taurus–A*

The position of this source closely coincides with that of the Crab Nebula (Plate 1), the expanding shell of an old supernova explosion. This object appears to be a likely source of high level radio-frequency noise. For a source size of five minutes the intensity at 100 Mc/s. corresponds to an equivalent temperature of about 2,500,000 °K. The present estimates of density and temperature in the Crab Nebula fall well short of explaining this result by strictly thermal processes. Non-thermal components resulting from turbulence within the nebula do not, however, seem unlikely.

(*d*) *Centaurus–A*

This source has been provisionally identified with the extra-galactic nebula N.G.C. 5128 (Plate 1). The fact that features of this nebula—the gas emission spectrum, the high density of interstellar matter, and the anomalous position of the obscuring band—are already known to be unusual suggests the possibility that its radio-frequency emission may also be unusual, and hence lends some support to the identification. (Recent work by Evans(9) raises the possibility that N.G.C. 5128 is not extra-galactic. He considers that this object is a gas cloud illuminated by a point at its centre. The dark marking across the nebula is presumably due to obscuring matter in the line of sight.)

(*e*) *Temporary Phenomena*

One example of a possible temporary source is 5·47 (probably in Centaurus, see Table 1). Some 10 records of this source at rising were obtained in November and December 1947. No further records were taken until June 1948, when no trace of the source could be found. Subsequent attempts to obtain records with higher sensitivity have also failed. There have been several other cases when small sources have been detected during systematic surveys—sometimes on several nights in succession—and attempts to secure additional confirmation at a later date have not succeeded. The possibility of receiver fluctuations accounting for such phenomena can be almost entirely excluded as the interference patterns obtained had lobe periods consistent with the bearing of the aerial. A much longer period of observation, however, will be necessary before the existence of temporary sources can be established beyond doubt.

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Fig. 1.—The nebula N.G.C. 5128.

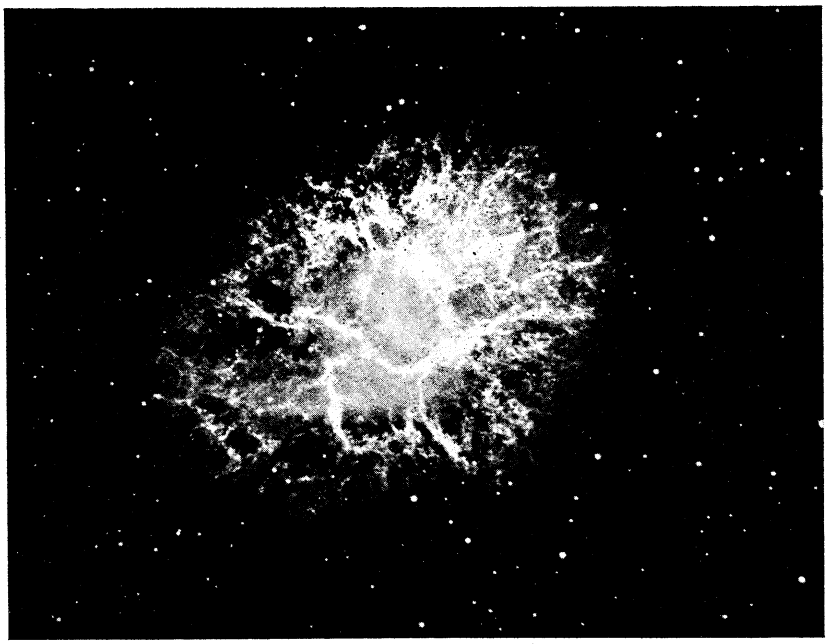


Fig. 2.—The Crab Nebula.

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(*f*) *Distribution of Sources*

The known sources are shown in Figure 3 on an equal-areas chart in galactic coordinates. The two dotted lines enclose the areas that are almost circumpolar to the present site, and thus have not been surveyed. The size of the dots gives a rough idea of the relative intensity of the sources. The number of sources available at present is too small to make any deductions from this distribution, but there seems a general tendency to crowd towards the galactic equator, and perhaps a noteworthy absence of sources between longitudes 330 and 210° south of the equator.

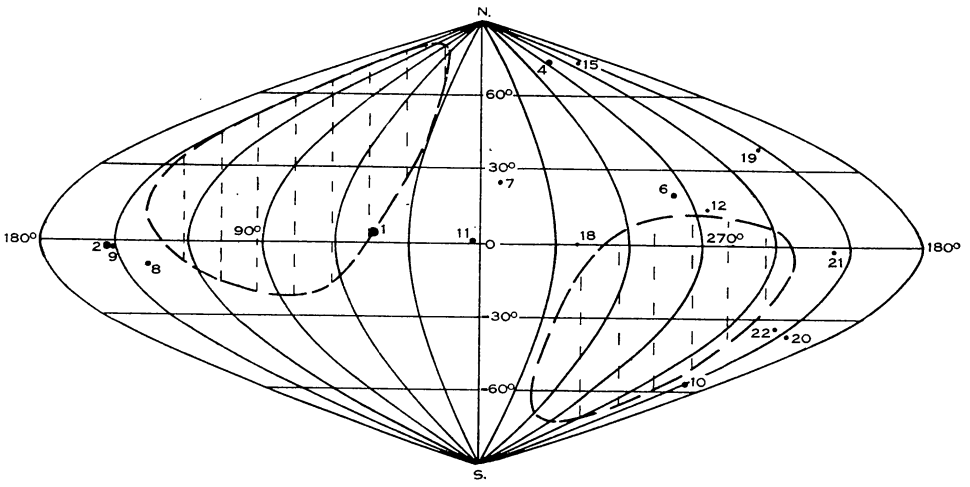


Fig. 3.—Distribution of discrete sources on an equal areas galactic chart. Numbers refer to Table 1.

IV. Noise Spectra

In Figures 4 (*a*) to 4 (*d*) noise " spectra " of the four most intense sources that can be observed from the present site are shown. Measurements were made on Cygnus and Taurus–A at five frequencies, 40, 60, 85, 100, and 160 Mc/s. The measurements on the sources Virgo–A and Centaurus–A were made during daylight when the terrestrial noise fluctuations were too great to obtain results at 40 Mc/s. A harmonic of a C.W. transmission prevented measurements of the intensity of Centaurus–A at 85 Mc/s.

Although the accuracy of these observations was not high, it was sufficient to show differences in the spectra of the various sources. For three of the sources the intensity is approximately inversely proportional to frequency, whereas for Taurus–A the intensity is apparently independent of frequency over the range of frequencies considered. The latter would be consistent with " thermal " radiation from an optically thin shell.

Figure 5 shows spectra of the north galactic pole and of a region round the galactic centre obtained with a single Yagi aerial. These are given in terms of equivalent temperature and the additional points at 40, 64, and 90 Mc/s. are due to Moxon(10) and Hey, Parsons, and Phillips(1). For the north galactic

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pole the equivalent temperature varies as $f^{-2.35}$ and for the galactic centre as $f^{-2.65}$ (in terms of intensity $f^{-0.35}$ and $f^{-0.65}$). It is interesting to note that, if a possible variation of source size with frequency can be ignored, for three of the discrete sources the intensity would increase more rapidly with decreasing frequency than that of the integrated background noise.

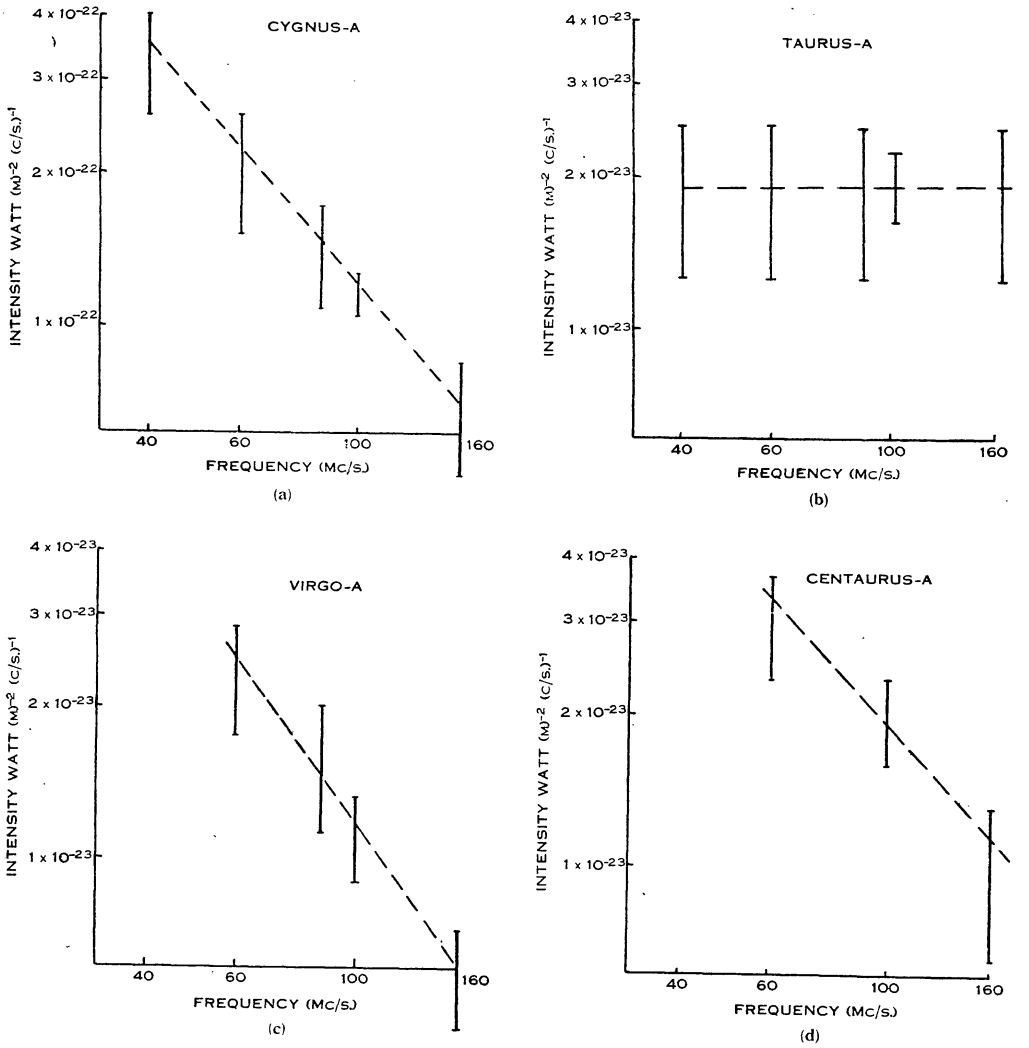


Fig. 4. (*a*) to (*d*)—Noise " spectra " of four discrete sources. Vertical lines indicate the range of observed values of intensity.

V. Fluctuations

Since their discovery by Hey, Parsons, and Phillips(1) the fluctuations in the received intensity of the source in Cygnus have remained one of the puzzling features of this phenomenon. It was at first thought that the radiation from the source consisted of two distinct components(2) as in the case of thermal and enhanced radiation from the sun. However, a longer period of observations

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has shown that this view is not correct, and that the mean intensity of the source is constant to within 10 per cent. Present knowledge of the fluctuations may be summarized as follows :

(i) The probability of fluctuations in intensity decreases with increasing radio frequency. Fluctuations have only rarely been observed at 160 Mc/s.

(ii) The amplitude (relative to the mean level) of the fluctuations increases with decreasing radio frequency.

(iii) The average period of the fluctuations decreases with decreasing radio frequency.

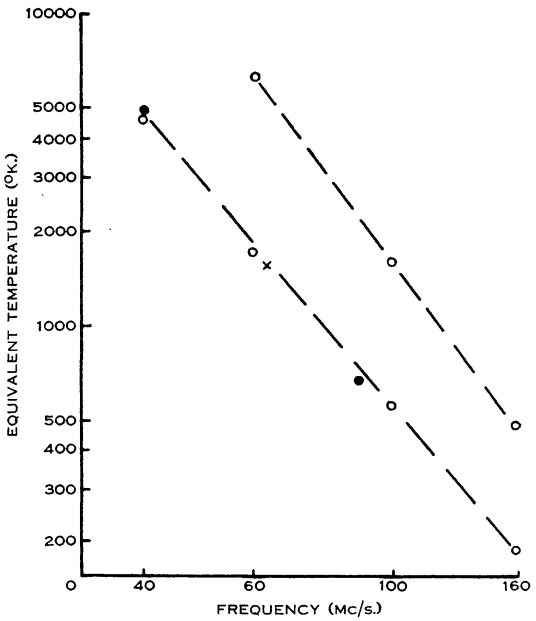


Fig. 5.—Variation of aerial temperature with frequency for the north galactic pole (lower curve) and a region surrounding the galactic centre (upper curve). o Radiophysics observations ; ● Moxon ; × Hey.

(iv) There is general correspondence between the degree of fluctuation on a number of radio frequencies observed simultaneously from the same site.

(v) There is no detailed correspondence between fluctuations on frequencies as close as 85 and 100 Mc/s. observed simultaneously from the same site.

(vi) There is some evidence of an annual variation in the occurrence of the fluctuations. The data in Table 2 refer to observations on the source of approximately 1 hour's duration immediately after its rising. From each record the amplitude of the fluctuations is assessed as a percentage of the mean intensity. The monthly means of the values obtained are plotted in Figure 6. The maximum degree of fluctuation occurs

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between July and September when the source rises in the early evening, and the minimum in February and March when the source rises about sunrise. Observations, however, commencing before and finishing after sunrise showed that no appreciable change in the character of the fluctuations occurred after the sun rose.

TABLE 2

FLUCTUATIONS OF SOURCE IN CYGNUS

Month	Mean Monthly Average of Amplitude of Fluctuations as Per Cent. of Mean Intensity	Number of Observations
1947		
July	60	10
December	55	10
1948		
January	42	8
February	—	—
March	17	12
April	32	21
May	51	23
June	56	23
July	61	23
August	49	11
September	41	26
October	52	18
November	58	22
December	58	10
1949		
January	62	5
February	20	3
March	11	16
April	32	7
May	51	15
June	44	17

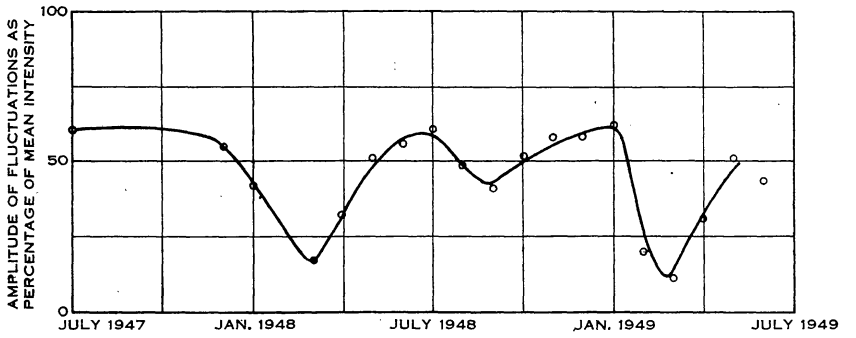


Fig. 6.—Degree of fluctuation of the intensity of the source in Cygnus. Individual points are monthly means.

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Records of the source taken simultaneously from sites on the east coasts of Australia and New Zealand showed no correlation in the fluctuations, whereas similar observations of enhanced solar radiation showed a 90 per cent. correlation. This indicates that the fluctuations are either extraordinarily directive or " locally " impressed.

Fluctuations in intensity have been observed for a number of the smaller sources, particularly Centaurus–A. The degree and incidence of fluctuation, however, is considerably less than that of the Cygnus source but is believed to follow the same annual curve. Fluctuations are most noticeable in all cases when the altitude of the source is less than 2°. It should be pointed out that observation of the very small sources depends on some degree of integration (Subsection II (*a*)) which would tend to obscure fluctuations.

There are four features of the fluctuation phenomenon which would suggest an origin in the ionosphere :

(i) The spaced aerial observations on Cygnus.

(ii) The annual variation in the Cygnus fluctuations.

(iii) The increasing degree of fluctuation with decreasing frequency.

(iv) Marked fluctuations in most cases at low altitude.

The absence of fluctuations in some of the minor sources may be due to angular width. If the fluctuations are due to irregularities in the ionosphere it is likely that the effect would be more pronounced in the case of sources of small angular width. Recent measurements indicate that the source in Cygnus has an angular width of less than 1′ 30″ whereas the angular widths of a number of the minor sources are believed to be several minutes of arc. This suggestion may be compared with the " twinkling " of stars and the steady appearance of planets.

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Appendix I

The Effect of Receiver Bandwidth on Interference Patterns

For the case of the sea interferometer described in Subsection II (*a*) the phase difference between the direct and reflected rays is $D+\pi$, where

$$D=\frac{4\pi h \sin \alpha}{\lambda}=\frac{4\pi hf \sin \alpha}{c} \quad \ldots\ldots\ldots\ldots (1)$$

If P_0 is the mean power in the band $\triangle f$ intercepted from the direct beam, the mean power intercepted from the two interfering beams in the elementary band df is

$$2P_0(1-\cos D)\frac{df}{\triangle f}.$$

The mean power over the band $\triangle f$, obtained by integrating between the limits $f\pm\frac{\triangle f}{2}$, is then

$$P=2P_0\left\{1-\cos D\frac{\sin\left(\frac{D\triangle f}{2f}\right)}{D\frac{\triangle f}{2f}}\right\}, \quad \ldots\ldots\ldots\ldots\ldots (2)$$

which reduces to the elementary case

$$P=2P_0(1-\cos D), \quad \ldots\ldots\ldots\ldots\ldots (3)$$

if $\triangle f$ is sufficiently small.

Under actual conditions $\left(\text{e.g. } \frac{\triangle f}{f}=\frac{1}{100}\right)$ the stationary values of P given by (2) are not sensibly different from those of (3). Thus minima occur where

$$D=2r\pi,$$

and maxima where

$$D=(2r+1)\pi \qquad r=0, 1, \quad \ldots\ldots\ldots\ldots\ldots (4)$$

Thus for the rth lobe

$$\frac{P_{min}}{2P_0}=1+\frac{\sin\left((r+\frac{1}{2})\pi\triangle f/f\right)}{(r+\frac{1}{2})\pi\triangle f/f}$$

and

$$\frac{P_{max}}{2P_0}=1-\frac{\sin\left(r\pi\triangle f/f\right)}{r\pi\triangle f/f}. \quad \ldots\ldots\ldots\ldots\ldots (5)$$

For the early lobes these values approximate to those of the elementary case, but as r increases they both tend to unity. Figure 7 shows the variation of $\frac{P_{max}-P_{min}}{4P_0}$ for values $\lambda=3$ m., $h=300$ m., and $\triangle f=1$ Mc/s.

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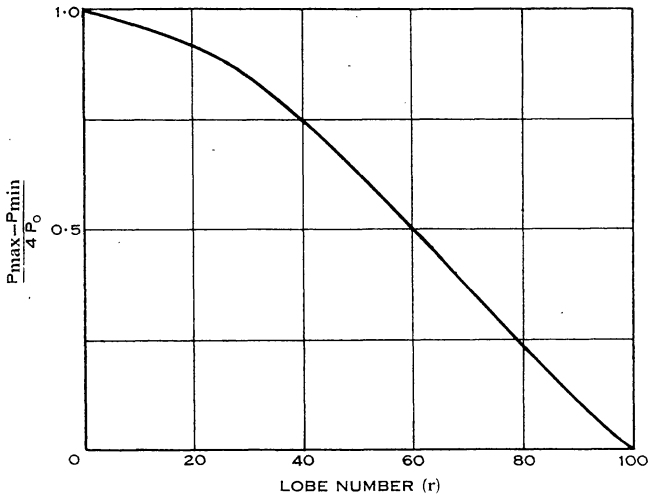


Fig. 7.—Illustration of the effect of receiver bandwidth on interference maxima and minima for numerical examples in Appendix I.

Appendix II

The Effect of Earth's Curvature on Sea Interferometer Patterns

The effect of the curvature of the earth is to reduce the power received by the aerial from the reflected beam (see Fig. 8). This results in incomplete interference.

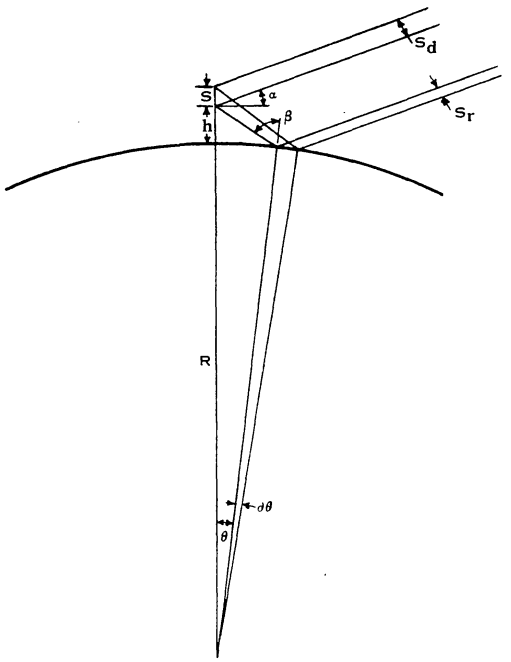


Fig. 8.—Effect of the earth's curvature on the reflected beam.

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If s is the " effective height " of the aerial, the widths of the direct and reflected beams intercepted from a source at altitude α are

$$s_\alpha = s \cos \alpha,$$

and

$$s_n = R\partial\theta \sin (\alpha+\theta), \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (1)$$

where $\partial\theta$ is the range of θ corresponding to the height s.

Since the effect occurs only at low altitudes the relation(2)

$$\alpha = \frac{h}{R\theta} - \frac{3\theta}{2} \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (2)$$

may be used.

Hence

$$R\alpha\theta = \frac{s}{\frac{h}{R\theta} + \frac{3\theta}{2}}$$

and (1) may be written

$$s_\alpha = s$$

$$s_n = s\,\frac{\frac{h}{R\theta} - \frac{\theta}{2}}{\frac{h}{R\theta} + \frac{3\theta}{2}}. \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (3)$$

The mean power intercepted from the interfering rays is then $2P_0(1-\cos D)\frac{s_r}{s}$ and the excess is $P_0\frac{(s_\alpha - s_n)}{s}$. The maximum and minimum values of P again occur where $D=2r\pi$ and $D=(2r+1)\pi$.

Then

$$\frac{P_{min}}{2P_0} = \frac{\theta}{\frac{h}{R\theta} + \frac{3\theta}{2}},$$

$$\frac{P_{max}}{2P_0} = \frac{2\,\frac{h}{R\theta}}{\frac{h}{R\theta} + \frac{3\theta}{2}} \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (4)$$

using the appropriate values of θ.

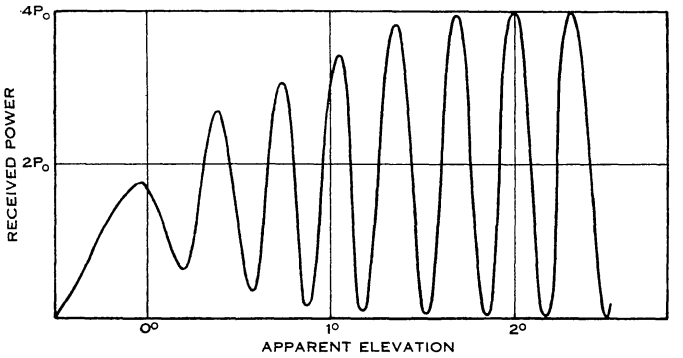


Fig. 9.—Theoretical interference pattern showing the effect of the earth's curvature on maxima and minima.

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Figure 9 is a computed interference pattern based on these formulae, taking $\lambda=3$ m., $h=300$ m., and neglecting the effect of receiver bandwidth. This pattern shows good agreement with actual patterns of the sources in Cygnus and Taurus obtained under the above conditions in New Zealand.

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